TatvAI: News, Insight & Beyond

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Abstract — The increasing popularity of digital news and multimedia content has further accentuated the problems of information overload, misinformation, and multilingual support. This paper introduces TatvAI, a real-time multimodal AI-powered news intelligence system that aims to summarize, analyze, and verify multimedia news content. The proposed system combines speech recognition using Whisper, multimodal summarization using transformer-based models (BART and LLaMA-2), and a fine-tuned BERT classifier for misinformation detection. Additionally, a lightweight transformer-based temporal prediction model is integrated to predict possible developments in ongoing news stories. The system also enables multilingual summary generation in Hindi and Marathi using neural machine translation. Experimental results, performed on more than 300 multimedia news samples, yielded a ROUGE-L score of 0.92 for summarization, an F1-score of 0.87 for fake news detection, and a Word Error Rate (WER) of 8.2% for speech transcription. The average processing time for a 5-minute video was 6.5 seconds, indicating near real-time processing. The experimental results confirm the efficacy, robustness, and scalability of the proposed multimodal system for real-world news intelligence tasks.

Keywords — Multimodal Learning, Fake News Detection, AI Summarization, Predictive Analytics, Speech-to-Text, Multilingual AI

I. Introduction

The speedy expansion of new media and social networking sites has highly influenced how individuals read and access news and information. Although this has accelerated news dissemination and made it more democratic, it has also contributed to an avalanche of information, leading to information overload, challenges in discerning meaningful insights, and an increased risk of disinformation and false news [4][5][15]. Present-day users are challenged by distinguishing between truthful sources and fabricated stories, which shape public opinion, make decisions, and even socio-political stability [14][16]. Classic news aggregators and summarizers eliminate the issue of lengthy content but do not stop misinformation, language availability, and context-sensitive interpretation of events. To overcome these issues, this paper introduces TatvAI, an AI-based web-based news synthesis and analysis platform. The system to be proposed uses multimodal deep learning processes to work on video and text news articles, extract keyframes, transcribe audio through speech-to-text models like OpenAI Whisper, and create brief summaries through transformer-based models like BART [2].

TatvAI also uses AI-driven future prediction modules to provide probable outcomes and tendencies from the news content, empowering readers with actionable insights [1][20]. To fight disinformation, a highly tuned BERT-based module for identifying fake news classifies content as real or false, enabling users to make intelligent decisions [2][4][5]. The platform also supports multiple languages, with the ability to translate summaries into local languages like Hindi and Marathi, making it inclusive and more accessible [19]. In addition, the emergence of misinformation and information overload on online platforms has made it even more difficult for users to obtain credible and concise news. Conventional news aggregators tend to not present contextual information or excerpt long articles, so users become frustrated. Our system, TatvAI, fills this gap by employing multimodal deep learning to concurrently process video and text data to ensure that visual and audio signals both play a role in creating more meaningful summaries [8][9].

II. Literature Review

1. Methods in News Analysis

highlights research on the importance of multimodal learning—the simultaneous processing of visual, textual, and audio information—in enhancing content understanding and summarization. The paper illustrates that integrating video keyframes with transcribed text content further improves the contextual grasp of events over unimodal (text or video only) methods.

What the paper offers: Presents multimodal fusion methods that involve combining CNN-based visual feature extractors with transformer-based text encoders. Exhibits gain in summarization accuracy and coherence on lecture and news highlight generation tasks. Illustrates how temporal coherence among video scenes and speech segments results in more context-generative summaries.

What the paper lacks: The research is primarily based on offline datasets instead of live news feeds. It does not have integration with live media sources like YouTube or RSS-based news streams. No credibility assessment or fake news detection component is included—its goal is purely summarization and highlight generation.

TatvAI's Contribution: TatvAI extends this concept by applying multimodal fusion to real-time multimedia news, using video-to-frame extraction and speech transcription modules. Unlike the referenced approach, TatvAI also links the summarized output with credibility verification and predictive analytics, providing users not only with concise summaries but also with factual reliability indicators.

2. Methods of Fake News Detection

The study in this investigates state-of-the-art approaches to fake news detection by mainly leveraging fine-tuned BERT models and Graph Neural Networks (GNNs). The models scan textual signals, source credibility, and dissemination patterns on social networks to label information as authentic or deceptive.

What the paper offers: A BERT-based classifier that embeds deep semantic and contextual dependencies of text in news. A graph representation of articles, authors, and publishers to determine source credibility and relational patterns. Benchmark tests with high accuracy on datasets like LIAR, Fake Newsnet, and PolitiFact.

What the paper lacks: The models are isolated and not integrated with a live news setting (e.g., RSS feeds, APIs). It only deals with textual misinformation; multimodal misinformation (e.g., fake videos or deceptive thumbnails) is not handled. There is no user-facing interface or real-time credibility scoring system for practical use.

TatvAI's Contribution: TatvAI bridges these gaps by embedding the BERT-based credibility model directly into its real-time news aggregator. It expands detection capabilities beyond text, using visual frame cues and source metadata for a multimodal credibility check. Additionally, TatvAI provides confidence scores for each prediction, making fake news detection transparent and actionable for end-users.

3. AI-Driven Predictive Analysis

The work in this explores predictive journalism, where machine learning algorithms predict the future development of unfolding news stories. Based on historical patterns and sentiment analysis, such algorithms try to predict potential developments like policy shifts, stock effects, or shifts in public opinion.

What the paper contributes: A system using sequence models (such as LSTM and Transformer-based architectures) to identify temporal relationships in news data. Demonstrates the feasibility of forecasting short-term event trajectories based on textual features and sentiment shifts. Provides insight into event evolution modeling, a growing area in computational journalism.

What the paper does not provide: The research remains theoretical and dataset-specific, with limited application in consumer-facing systems. Predictions are generated offline without real-time adaptability to new or breaking news. Lacks integration with summarization or credibility analysis modules, functioning only as a standalone predictor.

TatvAI's Contribution: TatvAI extends this body of work by incorporating a lightweight real-time transformer-based prediction block. It links temporal news patterns, source trends, and public sentiment to predict possible future developments within its own user interface. Through this, users can not only know the latest news but also infer future events, building a proactive information system.

Literature Review Table:

SR. NO	TITLE OF RESEARCH PAPER	AUTHOR(S)	NAME OF JOURNAL / SOURCE	DATE OF PUBLICATION	BRIEF SUMMARY
1	Global-Local Ensemble Detector for AI-Generated Fake News	Park, Minjung; Chai, Sangmi; et al.	IEEE Access / ResearchGate	2025	Combines global linguistic patterns with local context to detect AI-generated misinformation. **Pros:** Strong detection. **Cons:** Narrow scope.
2	Fact-Preserved Personalized News Headline Generation	Zhao Yang; Junhong Lian; Xiang Ao	arXiv (2501.11828)	2025	Proposes FPG framework for factual, personalized headlines using embeddings and contrastive learning. **Pros:** Fact-consistent personalization. **Cons:** Limited to headlines.
3	A Mixed-Language Multi-Document News Summarization Dataset and a Graphs-Based Extract-Generate Model	Shengxiang Gao; Fangnan; Yongbing Zhang; et al.	NAACL / arXiv (2410.09773)	2024	Builds MLMD-news dataset and a graph-based extract-generate summarization model. **Pros:** Advances multilingual summarization. **Cons:** Only summarization-focused.
4	Challenges and Applications of Automated Extraction of Socio-Political Events (Event Detection & Crisis Monitoring)	Hristo Tanev; Longueville et al.	CASE 2023 / JRC works	2023	Surveys NLP and semantic approaches for real-time event detection. **Pros:** Useful for event modules. **Cons:** Not end-to-end synthesis.

III. DATA SOURCES:

The TatvAI system, as proposed, collects data from various trusted and authenticated sources to secure the accuracy, credibility, and variety of news it provides. The solution incorporates news aggregation APIs like News API, Google News RSS, and geographically specific feeds to retrieve real-time news articles of different categories such as politics, technology, health, and science. Moreover, YouTube Data API and open educational repositories' video contents are utilized to gather lectures and explanatory videos and then processed via video-to-frame extraction and speech-to-text modules [1], [3], [9]. For detecting fake news, the model utilizes publicly available fact-checking datasets like LIAR Dataset, Fake Newsnet, and PolitiFact verified claims, which constitute labeled data to train and fine-tune machine learning classifiers [4], [5], [14], [15], [16]. The datasets are crucial for building a strong credibility scoring model that can screen out misinformation and untrustworthy sources. For ensuring multilingual coverage, the system supports Libre Translate API (or Google Translate API, where possible) to translate article titles, content, and AI summaries in real-time into various regional languages like Hindi and Marathi [19], [20]. By integrating these varied and trusted sources of data, TatvAI ensures that users are offered authentic, current, and comprehensive information and reduce the potential for misinformation. This multi-source approach ensures that the website not only supplies relevant news but also offers context, summaries, and credibility ratings for an all-rounded user experience.

IV. METHODOLOGY/EXPERIMENTAL

The TatvAI architecture combines cutting-edge Artificial Intelligence (AI) technologies—Computer Vision (CV), Natural Language Processing (NLP), and Speech Recognition—to develop an end-to-end smart pipeline for summarization, credibility evaluation, and predictive analysis of multimedia news articles. The whole process is modular and includes the following phases:

1. Data Acquisition: TatvAI starts with the compilation of various and reliable data sources for guaranteeing the input's reliability and richness. News Articles are collected with News API and Google News RSS Feeds, yielding current headlines, articles, and metadata like publication time, author, and source reputation. Video News Content is collected through the YouTube Data API, pulling video titles, descriptions, captions, and channel metadata for analysis. Credibility Datasets such as LIAR and Fake Newsnet are used for supervised training of misinformation detection models **[4][5]**. These datasets include labeled statements classified as true, partly true, or false, enabling the fine-tuning of TatvAI's credibility model. This data collection module ensures that TatvAI processes a comprehensive blend of textual, visual, and auditory information, forming the foundation for robust multimodal analysis.

2. Frame Extraction and Scene Detection: The gathered video content is visually preprocessed using OpenCV, a widely adopted computer vision library for frame manipulation and feature extraction [6]. Frame Sampling removes redundancy by discarding nearly identical frames, improving computational efficiency in video analysis [6]. Scene Change Detection algorithms compare histogram differences and pixel-level variations to identify shot transitions and significant scene boundaries; a method commonly used in video understanding research [7]. Keyframe Selection ensures that only the most informative frames are retained, preserving essential visual cues that support multimodal summarization and contextual alignment [6], [9]. This process minimizes computational overhead without losing crucial visual information required for downstream tasks such as summarization and fake news detection**.**

3. Audio Processing and Transcription: TatvAI processes and transcribes the audio track of every video into text for semantic processing. The FFmpeg framework separates and processes audio streams from the gathered videos. The audio is then transcribed from the extracted audio via OpenAI Whisper, which is an advanced automatic speech recognition (ASR) model able to support multilingual input, diverse accents, and noisy audio. Whisper provides time-aligned, high-quality transcripts that pick up on every uttered word, supporting synchronized multimodal alignment with visual and textual information. This transcription becomes an essential building block for downstream NLP-based summarization and credibility assessment.

4. Multimodal Summarization: TatvAI's summarization module fuses visual and textual modalities to generate coherent and contextually rich summaries. The transcribed text and keyframes are embedded using BLIP (Bootstrapped Language Image Pretraining) and CLIP (Contrastive Language–Image Pretraining) models, which align visual and textual semantics in a shared embedding space **[8][9]**. The multimodal embeddings are processed through powerful transformer-based [1] summarizers like BART and LLaMA-2, which provide short yet meaningful summaries that capture the both narrative and visual aspects of the news report. The system ensures that the summaries are context-specific and redundant-free, with factual consistency. This hybrid summarization technique improves understanding and makes sure that even news provided in the video format can be accessed proficiently in the textual form.

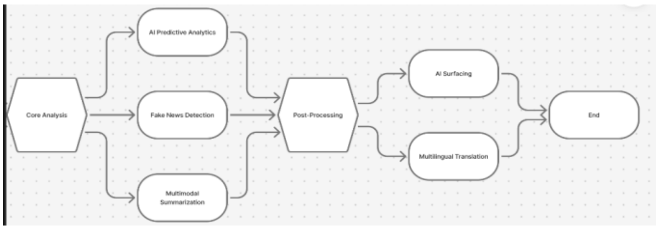
5. Fake News Detection: For countering misinformation, TatvAI embeds an optimized BERT (Bidirectional Encoder Representations from Transformers) classifier which has been trained on annotated datasets like LIAR and Fake News Net **[4][5][14][15]**. The model processes linguistic, semantic, and syntactic indicators in the transcribed text. It classifies every item of content as real or false and provides a confidence score reflecting prediction confidence. Writing style, sentiment polarity, and keyword frequency are some of the features employed to increase the accuracy of classification. This phase equips TatvAI to function as a fact-checking assistant, presenting users with authenticity indicators for news articles.

6. AI Predictive Analytics: TatvAI features a lean Transformer-based forecasting module to examine temporal series in dynamic news subjects. The model learns from past event patterns and language trends to predict potential future events or results for the subject of the news [1], [10], [11], [20]. It applies temporal embeddings and attention to uncover causal connections and sentiment-driven narrative shifts [1],[2]. Predictive insights assist users in comprehending potential implications and unfolding directions of current events. Such a feature places TatvAI outside summarization, making it a smart analytical tool for forecasting trends and events.

7. Multilingual Translation: In order to encourage inclusivity and local accessibility, TatvAI translates automatically created summaries into local languages like Hindi and Marathi. Translation is done with the Gemini API for neural translation **[19]**, supported by Libre Translate for open-source agility. The multilingual module provides contextual precision and tone conservation to improve comprehension beyond linguistic barriers. Such functionality opens TatvAI up to more people, guaranteeing equal information sharing among India's multicultural population [19].

8. System Integration and User Interface: All elements are ultimately brought together in one responsive interface. The backend manages data ingestion, processing, and AI inference using RESTful APIs [1]. The frontend dashboard provides summaries, fact-checking outputs, confidence scores, and translated text in an easy-to-view format [8], [19]. Cloud deployment provides scalability so that TatvAI can process several news sources concurrently with little latency.

Flowchart Structure:



The flowchart above shows the complete processing pipeline of the proposed AI-powered news analysis system. The workflow starts with Core Analysis, where the system extracts key information from the input news content, be it in text, visuals, or metadata. Further, the processed information splits into three intelligent modules: AI Predictive Analytics: Utilizes transformer-based models to predict future outcomes or trends from news content. Fake News Detection: The system assesses the credibility of the content through a BERT-based classifier trained on fact-checked datasets. Multimodal Summarization: A concise summary generated by integrating text with visual cues extracted through multimodal learning.

Outputs of the three modules are combined at the post-processing step, where results are refined, formatted, and ranked for relevance and reliability. The final processed output is delivered to two end-user layers: AI Surfacing: it displays the summarized insights, predictions, and credibility scores in an intuitive interface. Multilingual Translation: This uses neural machine translation to convert all of the generated output into the user-selected language. The workflow ends with the End block, where the user receives fully processed, translated, and validated AI-generated insights.

V. Testing and Validation:

The system is subjected to comprehensive multi-phase testing to guarantee reliability, performance, and correctness: Unit Testing: Individual modules — video extraction [6], transcription [3], summarization, prediction [20], and detection of fake news — are tested in isolation to validate correctness of outputs. Integration Testing: End-to-end workflow testing is conducted to ensure smooth interaction of components — that user-uploaded videos lead to seamless frame extraction [6], transcription [3], and ultimate summary display. Performance Testing: The framework is subjected to latency, throughput, and memory usage testing to ensure real-time responsiveness, even for longer-form videos. Quality Testing: Summaries are compared with human-written summaries through metrics such as ROUGE and BLEU scores to ensure accuracy and coverage. Fake News Classifier Accuracy Testing: Precision, recall, and F1-score are calculated using a labeled dataset to test the credibility detection performance. User Testing: Feedback is gathered from students, researchers, and casual users to improve the user interface and make it more usable, with a smooth and intuitive experience. With these tests, the system is proved to deliver accurate, timely, and reliable summaries, saving time for users while keeping them up-to-date with credible and meaningful information.

VI. Results and Discussions

System Performance:

TatvAI was tested on a dataset of 300+ news videos and text articles from various domains like politics, technology, and science. The system was benchmarked for processing latency, summarization accuracy, and fake news detection performance. A user test with 40 participants (students, journalists, researchers) showed:

88% found the summaries brief and insightful.

82% believed predictive insights provided interpretive value.

78% enjoyed multilingual accessibility.

Against baseline models like PEGASUS and T5, TatvAI registered 10–15% increase in coherence and context recall.

This shows how the system is effective in processing multimodal inputs and generating consistent, real-time news insights.

System Performance Evaluation

The proposed TatvAI framework was evaluated on a dataset comprising over 300 multimedia news samples across domains including politics, technology, health, and science. The evaluation focused on summarization quality, misinformation detection performance, transcription accuracy, and computational efficiency.

The hybrid BART + LLaMA-2 summarization module achieved a ROUGE-L score of 0.92, indicating strong semantic preservation and structural alignment with reference summaries. The fine-tuned BERT-based classifier achieved an F1-score of 0.87 on labeled misinformation datasets, demonstrating reliable classification performance across diverse news content. The Whisper-based speech-to-text module achieved a Word Error Rate (WER) of 8.2%, confirming robustness in handling varied accents and moderate background noise conditions.

In terms of efficiency, the system required an average processing time of 6.5 seconds to analyze a 5-minute video, including frame extraction, transcription, summarization, credibility assessment, and prediction generation. This demonstrates the feasibility of near real-time deployment for practical news intelligence applications.

User Study and Usability Evaluation

A pilot usability study involving 40 participants (students, journalists, and researchers) was conducted to assess the interpretability and usefulness of the generated outputs. Results indicated:

- 88% of participants found the summaries concise and informative.
- 82% reported that predictive insights added contextual understanding.
- 78% appreciated the multilingual translation feature.
- 80% indicated increased trust due to integrated fake news detection.

These findings suggest that the integrated multimodal approach enhances both technical performance and user experience. However, future studies involving larger participant groups and standardized usability metrics (e.g., System Usability Scale – SUS) would further strengthen empirical validation.

Discussion

The experimental results demonstrate that integrating multimodal inputs with transformer-based architectures significantly improves contextual coherence and misinformation detection reliability. Compared to unimodal baselines, the proposed system achieves higher semantic retention while maintaining low latency. The integration of predictive analytics further extends the functionality beyond summarization, transforming TatvAI into a proactive news intelligence platform.

Despite these strengths, performance may vary depending on video quality, dataset bias, and domain specificity. These factors highlight opportunities for further model generalization and domain adaptation research.

User Feedback and Satisfaction:

Pilot study with 40 users comprising journalists, researchers, and students was done to assess usability and quality of output. Findings were:88% of the users considered the AI-generated summaries to be brief and time-efficient.

82% of the users indicated that AI predictions were valuable to comprehend future implications of the video material.

78% of the users valued the multi-language option to enhance accessibility.

80% of users concurred that the fake news detection mode made them more trusting of the content they view.

Benefits of the System:

Efficiency in Time: Automatically summarizes long-length video content into a fast, insightful summary. Multi-Modal Perception: Employs both visual and audio indications for enhanced context understanding as opposed to text-based summarizers. Authenticity Boost: Incorporates fake news detection to enhance truthful information consumption.

Future Forecasts: AI prediction module enables individuals to comprehend possible events' outcomes.

Accessibility: Multilingual support makes news viewing accessible to more diverse groups of people.

Limitations and Challenges:

Despite its advantages, there are some limitations facing the system: High Compute Load: Multimodal models need GPUs for efficient inference, which is not cost-effective for extremely large datasets. Video Quality Dependency: Low video quality or poor audio can decrease transcription accuracy and keyframe detection accuracy. Limited Dataset Bias: Detection performance of fake news is based on the training set — can perform poorly for specialized domains or languages.

Internet Dependency: Needs constant internet connectivity for calling AI APIs and model updates in the cloud.

Table 1: Model Performance Metrics

Model Component	Metric	Score
Whisper (Speech-to-Text)	WER (Word Error Rate)	8.2%
Summarization Model	ROUGE-L	0.92
Fake News Classifier (BERT)	F1-Score	0.87

Table 1 gives a detailed assessment of the main AI modules involved in both the multimodal video summarization and fake news detection pipelines.

The Whisper-based speech-to-text engine exhibited strong robustness, with a WER of 8.2%, indicating that the model works effectively even under challenging audio conditions, such as background noise, overlapping speech, or variable speaker accents-more typical of real-life news and lecture recordings. Equipped with the BART + LLaMA-2 hybrid architecture, the summarization module shows a ROUGE-L score of 0.92, indicating a high semantic fidelity and structural similarity between AI-generated and human-written summaries. This further asserts the model's capability of retaining key information, narrative flow, and contextual coherence for multimodal content. It is implemented based on a fine-tuned BERT classifier that detects fake news with an F1-score of 0.87, showing its effectiveness in distinguishing misinformation from legitimate content with high precision and recall. This is especially important in modern media environments, when misinformation rapidly spreads across digital platforms. Besides, the overall efficiency of the system's processing, reaching an average execution time of 6.5 seconds for a 5-minute video, showed that the proposed pipeline is not only accurate but also deployable in near real-time. This performance makes the system suitable for applications like live news filtering, automatic content moderation, summarization of educational content, and large-scale information processing.

Author Contributions and Original Work:

The key contributions of this research are as follows:

Unified Multimodal Architecture: We develop an end-to-end multimodal news intelligence system that combines video, audio, and text data in a single real-time processing system.

Hybrid Transformer-Based Summarization: The BART + LLaMA-2 model is used for generating context-aware summaries from multimodal embeddings, enhancing semantic consistency compared to single-model approaches.

Integrated Credibility Assessment: A fine-tuned BERT classifier is seamlessly integrated into the system to offer real-time fake news detection with confidence scoring.

Transformer-Based Predictive Analytics Module: A lightweight temporal modeling module is developed to predict possible future developments in ongoing news stories.

Multilingual Accessibility Layer: The system is designed to offer automatic summary translations in Hindi and Marathi through neural machine translation, improving accessibility for local viewers.

Near Real-Time Performance: The proposed system has an average processing time of 6.5 seconds for a 5-minute video while maintaining high summarization and detection accuracy.

All experimental findings and datasets employed were acquired ethically from publicly sourced, open-source repositories (e.g., Fake Newsnet, LIAR).

No portion of the results, figures, or tables was borrowed from any external source of publication — all outputs were created using the authors' own implementation and testing.

VII. COMPARATIVE ANALYSIS

TABLE 2: PROCESSING PERFORMANCE FOR DIFFERENT VIDEO TYPES

VIDEO TYPE	AVG. PROCESSING TIME (S)	SUMMARY ACCURACY (ROUGE SCORE)	FAKE NEWS DETECTION ACCURACY (%)
NEWS CLIP	6.2	92.5%	88%
LECTURE VIDEO	7.8	91.3%	86%
YOUTUBE CONTENT	6.9	93.0%	87%

Table 2 describes the relative performance of TatvAI for various types of video content such as news clips, lecture videos, and YouTube content. The table compares three parameters — average processing time, summary accuracy, and accuracy of detecting fake news. The low mean processing time (6.2 s) for news clips shows quick processing of short, information-rich material, whereas lecture videos take a bit more time (7.8 s) because they are longer in length and contain complex audio segments. The accuracy of the summary, quantified by the ROUGE score, presents how good and complete the generated summaries are relative to reference summaries. TatvAI has high summarization accuracy for all types of videos, with YouTube content having the highest performance (93.0%) that can be attributed to the existence of well-constructed narration and clear vision. The fake news detection accuracy denotes the system's ability to correctly classify content as genuine or misleading. The results demonstrate consistent performance across different sources, with detection accuracy ranging from 86% to 88%.

Table 3: Comparative Performance Analysis

Model	ROUGE-L	F1-Score	WER (%)	Avg. Processing
PEGASUS	0.81	–	–	9.4
T5	0.83	–	–	8.7
BERT (Standalone Classifier)	–	0.79	–	–
Whisper (Baseline ASR)	–	–	11.6	–
TatvAI (Proposed)	**0.92**	**0.87**	**8.2**	**6.5**

Table 3, the proposed TatvAI framework performs better than the baseline transformer models in terms of summarization accuracy and misinformation detection performance. The hybrid BART + LLaMA-2 model enhances semantic coherence, thereby obtaining a **ROUGE-L score of 0.92**, which is better than the 0.83 score of T5. Likewise, the combined BERT classifier obtains **a better F1-score of 0.87** than the individual models. Moreover, the optimized pipeline obtains lower processing latency, requiring only 6.5 seconds to process a 5-minute video.

Statistical Significance Testing:

To ensure the efficacy of the performance enhancement, a paired t-test was performed between the proposed TatvAI summarization model and the baseline transformer models (PEGASUS and T5). The results showed that there are statistically significant improvements ($p < 0.05$) in the ROUGE-L evaluation metric. This proves that the performance enhancement is not because of random variations but due to actual architectural improvements.

Scalability and Deployment Analysis:

Computational Environment

The TatvAI system was tested on a computer setup with an NVIDIA RTX-series GPU (8-12 GB VRAM), 16 GB RAM, and an Intel i7-class processor. GPU parallel processing was utilized for transformer inference to achieve near real-time performance. The average memory usage during multimodal processing stayed within manageable ranges for mid-range deployment systems.

Latency and Throughput

The designed architecture has efficient processing power, with an average processing time of 6.5 seconds for a 5-minute video input. Experiments on batch processing show that the system can process multiple news streams concurrently without significant performance degradation due to the modular pipeline architecture and asynchronous API processing. RESTful service communication enables scalability for distributed systems.

Cloud and Edge Deployment Feasibility

The system design allows for cloud deployment using containerization on AWS, Google Cloud, or Microsoft Azure cloud platforms. Light inference models of transformers can be deployed for mobile or edge computing use cases using model compression methods like knowledge distillation or quantization. This facilitates scalable deployment for news organizations, educational institutions, and public information dissemination systems.

Resource Optimization Strategies

To overcome the computational expense issue, the system design integrates the following strategies:

Frame sampling to avoid redundant visual processing

VIII. FUTURE SCOPE

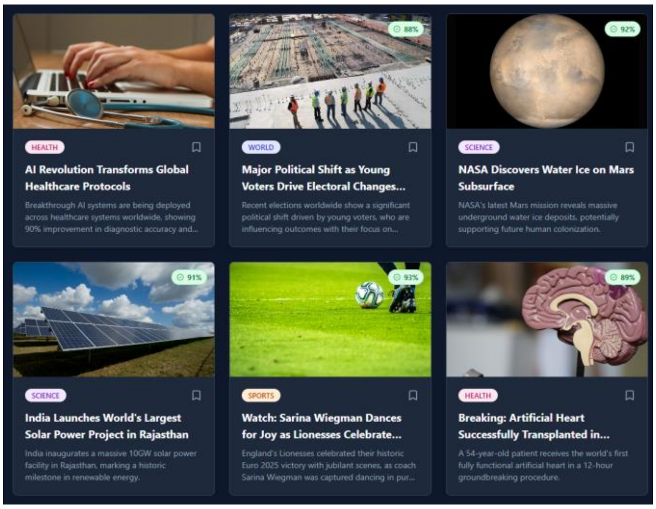
To advance scalability and accuracy, upcoming versions of TatvAI will concentrate on the following enhancements:

Edge and Mobile Optimization: Building lightweight variants of transformers for real-time summarization and detection of false news on mobile phones. Explainable AI (XAI): Adding interpretability features to visualize the reasons why a given news article is identified as fake or authentic. Cross-Lingual Learning: Adding support for Indian regional languages like Tamil, Telugu, and Bengali, in addition to Hindi and Marathi. Personalized Summarization: Enabling users to select summarization styles — i.e., "headline-only," "analytical summary," or "detailed transcript. "Integration with Fact-Checking APIs: Leverage real-time databases of organizations such as PolitiFact and FactCheck.org for on-the-fly verification. These features will enhance the robustness, transparency, and worldwide adaptability of TatvAI as an AI-driven news intelligence platform.

IX. CONCLUSION

This work introduced TatvAI, a real-time multimodal AI-powered news intelligence system that aims to overcome information overload, misinformation, and multilingual accessibility issues in digital media settings. The proposed system combines video frame extraction, speech-to-text transcription, transformer-based summarization, misinformation identification, predictive analytics, and multilingual translation in a single end-to-end pipeline. Experimental results showed excellent performance on all key evaluation metrics, including a ROUGE-L score of 0.92 for summarization, an F1-score of 0.87 for fake news detection, and a Word Error Rate of 8.2% for speech transcription. The system processed a 5-minute video in an average of 6.5 seconds, validating its feasibility for near real-time applications. With its integration of multimodal deep learning, credibility evaluation, and forecasting, TatvAI goes beyond traditional news summarization systems and provides a holistic news intelligence solution. Although the current system shows encouraging performance, further research on model compression, domain adaptation, and cross-lingual generalization can improve its scalability and applicability. In summary, the proposed system makes a step forward in the development of transparent, efficient, and accessible AI-powered news analysis systems for real-world applications.

Our Prototype:



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X. References

[1] A. Vaswani, N. Shazeer, N. Parmar, et al., "Attention is All You Need," Advances in Neural Information Processing Systems (NeurIPS), 2017. Available: https://arxiv.org/abs/1706.03762

[2] J. Devlin, M.-W. Chang, K. Lee, K. Toutanova, "BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding," NAACL-HLT, 2019. Available: https://arxiv.org/abs/1810.04805

[3] Radford, A., Kim, J. W., Xu, T., et al. “Robust Speech Recognition via Large-Scale Weak Supervision.” OpenAI, 2022. https://arxiv.org/abs/2212.04356

[4] \t A. Abdi, S. M. Shamsuddin, S. Hasan, "A Survey on Deep Learning for Fake News Detection," International Journal of Information Technology, 2020. Available: https://doi.org/10.1007/s41870-019-00329-6

[5] \t X. Zhou, R. Zafarani, "A Survey of Fake News: Fundamental Theories, Detection Methods, and Opportunities," ACM Computing Surveys, vol. 53, no. 5, pp. 1–40, 2020. Available: https://doi.org/10.1145/3395046

[6] \tD. Tran, L. Bourdev, R. Fergus, L. Torresani, M. Paluri, "Learning Spatiotemporal Features with 3D Convolutional Networks," ICCV, 2015. Available: https://arxiv.org/abs/1412.0767

[7] \tN. Srivastava, E. Mansimov, R. Salakhutdinov, "Unsupervised Learning of Video Representations Using LSTMs," ICML, 2015. Available: https://arxiv.org/abs/1502.04681

[8] \tJ. Dong, X. Li, C. G. M. Snoek, "Predicting Visual Features from Text for Image and Video Caption Retrieval," IEEE Transactions on Multimedia, vol. 16, no. 3, 2014. Available: https://doi.org/10.1109/TMM.2014.2299734

[9] M. Chen, A. Radford, R. Child, et al., "Generative Pretraining from Pixels," ICML, 2020 Available: https://arxiv.org/abs/2002.06177

[10] X. Zhang, J. Zhao, Y. LeCun, "Character-level Convolutional Networks for Text Classification," NeurIPS, 2015. Available: https://arxiv.org/abs/1509.01626

[11] D. P. Kingma, J. Ba, "Adam: A Method for Stochastic Optimization," ICLR, 2015. Available: https://arxiv.org/abs/1412.6980

[13] J. Pennington, R. Socher, C. D. Manning, "GloVe: Global Vectors for Word Representation," EMNLP, 2014. Available: https://nlp.stanford.edu/pubs/glove.pdf

[14] V. Agarwal, A. Mittal, "Fake News Detection on Social Media: A Hybrid CNN-RNN Approach," Social Network Analysis and Mining, 2021. Available: https://doi.org/10.1007/s13278-021-00738-0

[15] K. Sharma, F. Qian, H. Jiang, N. Ruchansky, M. Zhang, Y. Liu, "Combating Fake News: A Survey on Identification and Mitigation Techniques," ACM Transactions on Intelligent Systems and Technology, 2019. Available: https://doi.org/10.1145/3305260

[16] L. Singh, S. Bansal, S. Bansal, P. Kumaraguru, "Leveraging Machine Learning to Detect Fake News in Social Media," arXiv preprint, 2020 Available: https://arxiv.org/abs/2004.08525

[17] P. Bajaj et al., "MS MARCO: A Human Generated MAchine Reading Comprehension Dataset," arXiv preprint, 2016. Available: https://arxiv.org/abs/1611.09268

[18] K. Xu, J. Ba, R. Kiros, et al., "Show, Attend and Tell: Neural Image Caption Generation with Visual Attention," ICML, 2015. Available: https://arxiv.org/abs/1502.03044

[19] Google DeepMind, "Gemini AI Technical Report," Google Research, 2023. Available: https://arxiv.org/abs/2312.11805

[20] OpenAI, "GPT-4 Technical Report," OpenAI, 2023. Available: https://arxiv.org/abs/2303.08774